Shagun Ravi Khemka

Senior / Lead Backend Engineer · AI Systems | Mumbai, India
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Senior Software Engineer with **7+ years** building high-scale distributed systems and **production LLM systems**. At JioHotstar, driving **280M+ user subscription** growth end-to-end, from architecture to cross-functional delivery, on a platform processing **50,000 RPS** at peak.

Technical Skills:

- **Distributed Systems & Infrastructure:** High-scale microservices, event-driven architecture, system design, REST API design, gRPC, performance optimization, capacity planning.
- **Data & Databases:** Relational (PostgreSQL, MySQL), NoSQL (DynamoDB, Google Spanner), Caching (Redis), Stream processing (Apache Kafka, Apache Flink).
- **Cloud & DevOps:** AWS/GCP, Kubernetes, Docker, CI/CD, Observability (Prometheus, Grafana), on-call rotations.
- **AI Engineering:** LLM model routing, eval harnesses, guardrails, structured outputs, prompt caching, cost/latency observability; agentic coding (Claude & Codex skills).
- **Leadership & Strategy:** Technical roadmap design, architectural decision-making, team leadership (4+ engineers), mentorship, cross-functional project leadership, incident management, technical interviews.
- **Languages:** Go (Golang), Python, Java, TypeScript.

Experience

JioHotstar – *Senior Software Engineer* - Backend - Bangalore, India **March 2024 - Present**

- Re-engineered legacy subscription/upgrade flows in **Java (Spring Boot, JPA)**, decoupling purchase logic into a unified, extensible architecture that drove **~20% revenue growth** and **~40% higher paywall** conversion at **2,000 TPS** peak. **(Recognised with Astral Team Award, Q4 FY25–26)**
- Architected and led the Subscriptions Read P0 service serving all internal read APIs (token, entitlement checks for Identity and Playback) at 3**M RPM peak (~50K RPS)**; held strict **P99 SLAs** while running **~20% fewer pods**, cutting infra cost with no perf regression.
- Led cross-functional migration of **14M+ users** from JioCinema to Disney+ Hotstar, ensuring consistency across subscription state, payment reconciliation, and entitlement systems.
- Supported production on-call and IPL live events at **300M+ views / 30M+ peak concurrency**, ensuring platform stability and rapid incident resolution.
- Migrated a legacy Java service to **AWS Graviton (ARM)** + **Java 21 (~20% compute savings),** codifying the playbook into reusable Claude & Codex AI-agent skills for team-wide self-serve migrations.

FamPay(YC S19) – *Software Engineer - Backend* - Bangalore, India **June 2020 - March 2024**

- Led a team of 4 backend engineers, owned sprint planning, architectural decisions and code reviews while driving architecture, scalability, and operational excellence for the FamCard microservice; supported **5M+** virtual cards, **200K+** physical deliveries, **₹10M+** in daily transaction volume, and achieved a **~30%** reduction in card delivery costs.
- Spearheaded revenue-generating features for **5M+** monthly transacting users (MTUs), including Recharge, Gift Cards, and Electricity, contributing to ~**60%** of the company's total revenue.
- Supported Leadership & Talent in **hiring**, by conducting technical interviews and mentoring 5+ backend interns and junior developers who secured full-time roles at Fam and leading tech companies.

Instamojo – *Software Engineer* - Backend - Bangalore, India **January 2019 - May 2020**

- Automated the merchant onboarding flow, reducing KYC resubmissions by **~25%**, and earned a **full-time offer (PPO)** based on performance during the internship.
- Built an ingestion service for the reseller program, enhancing product visibility and search; enabled **~15K** products to be onboarded within the first month of beta launch.

Education

VIT Vellore - Bachelor in Information Technology (CGPA - 8.32/10) 2015 - 2019

IIIT Bengaluru - PG Diploma in Software Development (Blockchain) (CGPA - 3.73/4) 2019 - 2020